welcome

Thanks for purchasing the MEAP for *Algorithms and Data Structures in Action*. This book is meant to help developers demystify algorithms, develop the ability to identify performance criticalities and improve the quality and speed of code.

Algorithms and data structures are the hearth of software and hardware, they are pervasive, even if we don’t realize it. They handle everything from low-level tasks on chips, like divisions or square roots, to complex tasks on different chips, like reconstructing a sampled audio signals using discrete Fourier transform on your smartphone, to even more complex tasks on the cloud, as in scheduling the allocation of resources on a Kubernetes master node.

It would be impossible to gather a comprehensive catalog of all existing algorithms, many of them are designed to run in a very narrow context, and even just considering the general purpose ones, there are so many that no single engineer, computer scientist or part-time-genius could possibly remember them all; the goal for the interested reader should be, instead, to build awareness and familiarity with increasingly complex algorithmic concepts, developing the ability to look at problems with a critical eye and spot opportunities to make code or design more efficient by using the right data structure.